

Standard Data Products

APPN Aerial SOP — Locked revision 1.0

Standard Data Products

This page provides an overview of the **standard data products** generated by APPN UAV processing pipelines, including typical spatial resolution, file formats, and software compatibility. By default, products are delivered in **open or widely supported formats** that can be accessed using **freely available tools**.

For the pipelines that generate these products, see Processing Pipelines.

Guiding principles

- Products are generated using standardised, documented pipelines to ensure consistency across platforms and sites.
- Output formats prioritise accessibility using freely available and open-source software.
- Spatial resolution and product definitions are explicit and traceable.
- Products are intended to be suitable for downstream analysis, comparison, and reuse.

Standard LiDAR-derived products

Product	Description	Typical resolution	Typical file size (<i>indicative</i>)
LiDAR Digital Surface Model (DSM)	Surface model representing canopy and above-ground features	8 cm	Derived from LiDAR point cloud
LiDAR Digital Terrain Model (DTM)	Ground surface model with vegetation removed	1 m	Derived from LiDAR point cloud
Combined LiDAR Point Cloud	Aggregated and filtered point cloud	Native point spacing	~110–175 MB per minute of flight

Typical LiDAR file sizes are based on raw LiDAR logging rates documented in the CALViS and GOBI SOPs; DSM and DTM sizes scale with survey area and output resolution.

Standard hyperspectral products

Product	Description	Typical resolution	Typical file size (<i>indicative</i>)	Format
VNIR Orthomosaic	Radiometrically calibrated VNIR hyperspectral mosaic	~4 cm (GSD-based)	Several GB per flight (even for short missions)	ENVISAT
SWIR Orthomosaic	Radiometrically calibrated SWIR hyperspectral mosaic	~4 cm (GSD-based)	Several GB per flight (even for short missions)	ENVISAT

Hyperspectral data volumes scale strongly with mission duration, frame period, and number of spectral bands. SOPs note that even short flights routinely generate multi-GB datasets.

Standard RGB products

Product	Description	Typical resolution	Typical file size (<i>indicative</i>)	Format
RGB Orthomosaic	High-resolution RGB orthorectified mosaic	~0.6 cm (fixed)	~30–70 MB per image (image every ~2 s by default)	GeoTIFF (.tif)